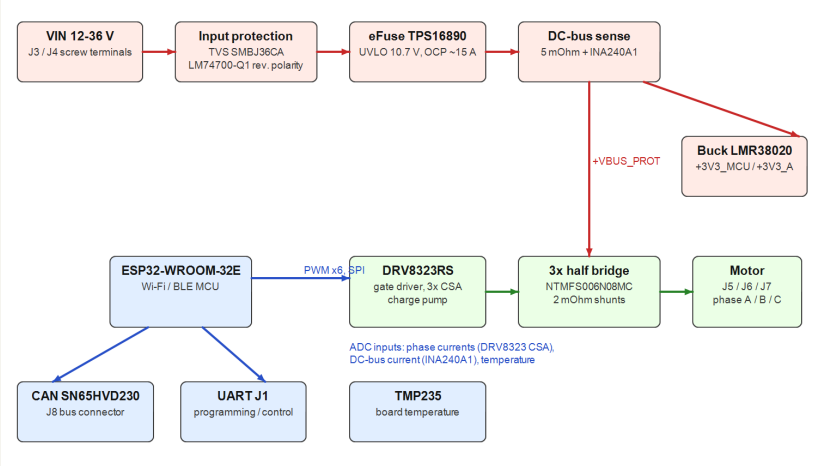
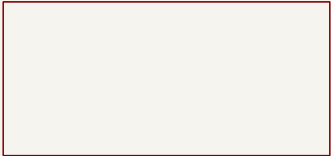


3-Phase BLDC / PMSM Electronic Speed Controller
Revision B – 2026-09-27

Input voltage: 12-36 V DC (TVS SMBJ36CA, reverse polarity LM74700-Q1, eFuse TPS16890: UVLO ~10.7 V, circuit breaker ~15 A)
Phase current: 5-10 A continuous (NTMFS006N08MC 80 V, 2 mOhm shunts)
Gate driver: DRV8323RS (SPI, 3x current-sense amplifiers)
MCU: ESP32-WROOM-32E, soldered directly (Wi-Fi/BLE)
Interfaces: UART (programming + control), CAN 2.0 (SN65HVD230)
Sensing: 3x phase current, DC-bus current (INA240A1), board temperature
Logic supply: LMR38020 buck, +3V3_MCU / +3V3_A
PCB: 2 layers, 1 oz, 1.6 mm FR-4, PCBWay standard spec

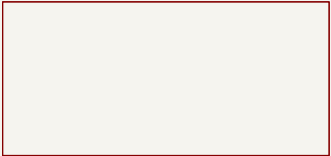


Power Input & Protection



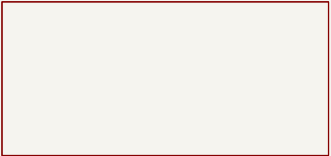
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Power Supply 3.3 V



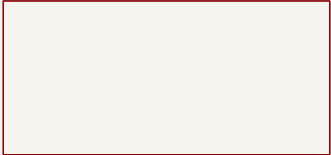
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DC-Bus Current Sensing



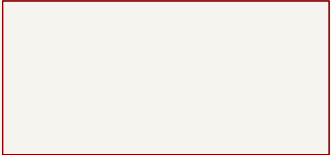
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MCU Core



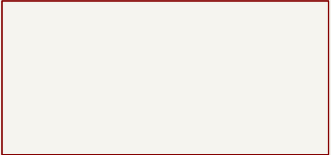
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Gate Drive



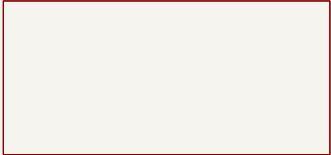
File: GateDrive.kicad_sch

Power Stage



File: PowerStage.kicad_sch

Communication



File: Communication.kicad_sch

Revision history
A 2026-09-27 First version – withdrawn (schematic net shorts, rule violations)
B 2026-09-27 Full rebuild per METU PowerLab PCB design rules

Title: 3-Phase BLDC ESC – Overview

Project: 3-Phase BLDC ESC

Number: ESC3PHASE-001

Rev: B

Sheet: /

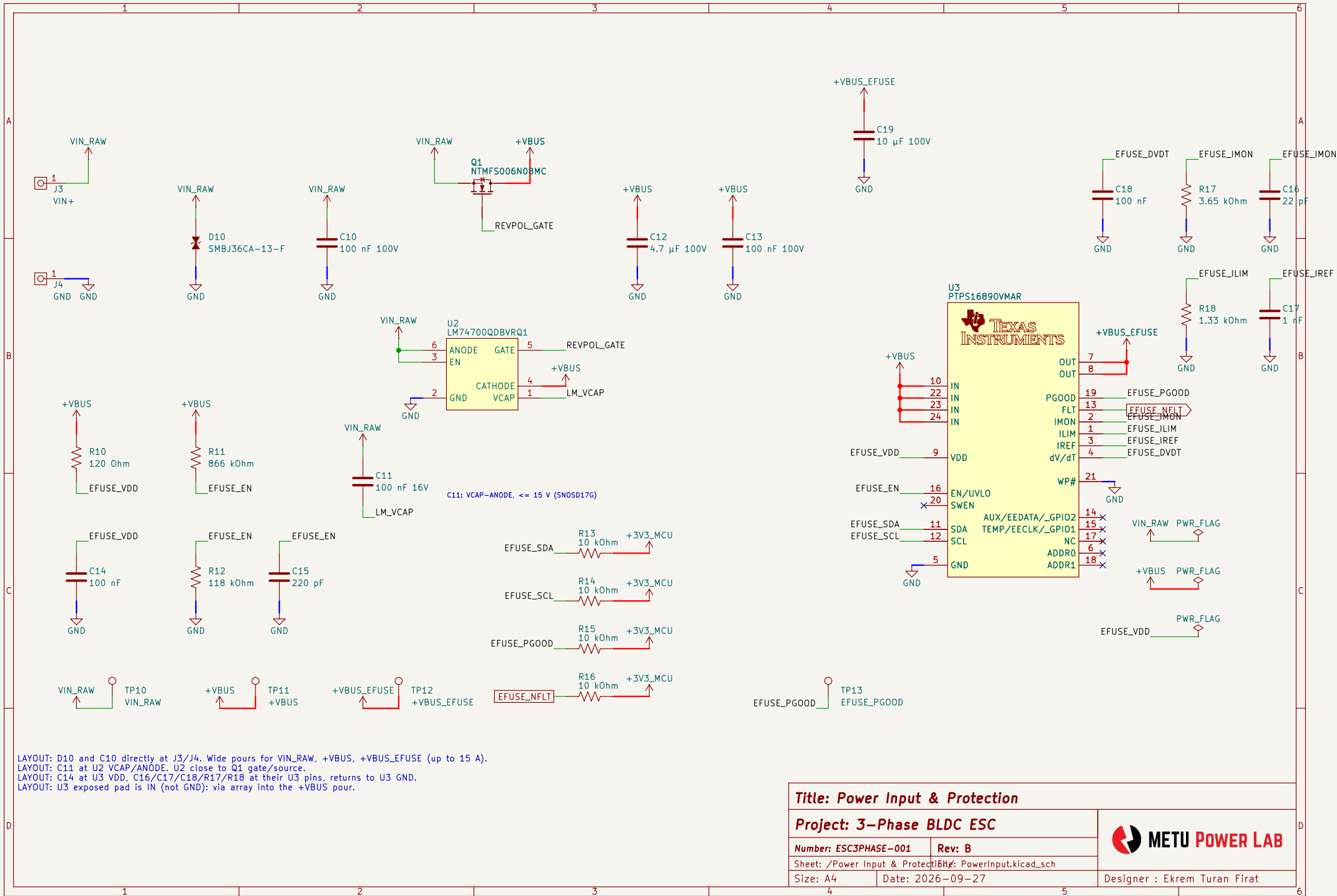
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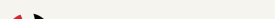
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Date: 2026-09-27

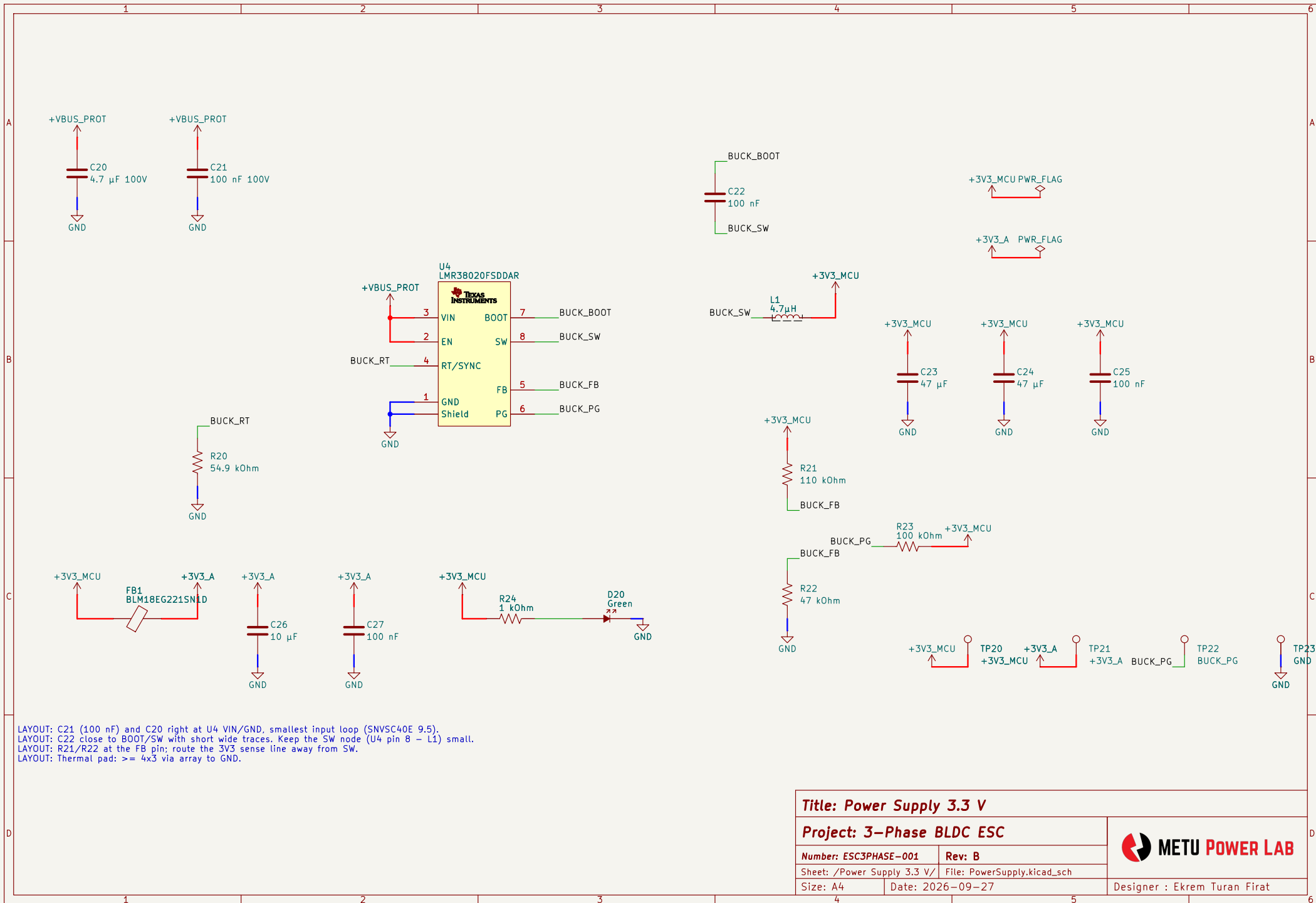


Designer : Ekrem Turan Firat



Title: Power Input & Protection		
Project: 3-Phase BLDC ESC		
Number: ESC3PHASE-001	Rev: B	
Sheet: /Power Input & Protection.kicad_sch		
Size: A4	Date: 2026-09-27	
		Designer : Ekrem Turan Firat





Title: Power Supply 3.3 V

Project: 3-Phase BLDC ESC

Number: ESC3PHASE-001

Rev: B

Sheet: /Power Supply 3.3 V/

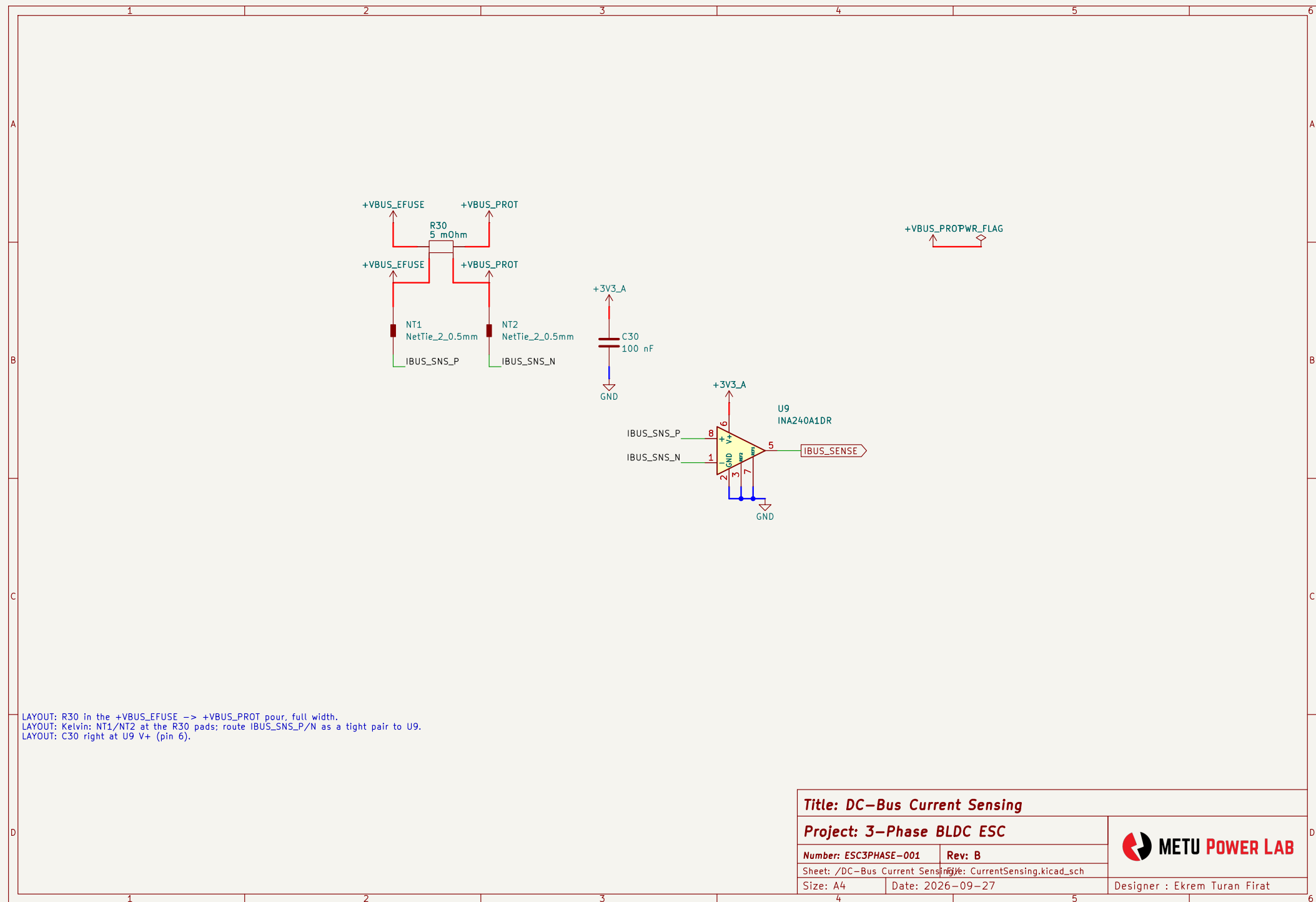
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Date: 2026-09-27

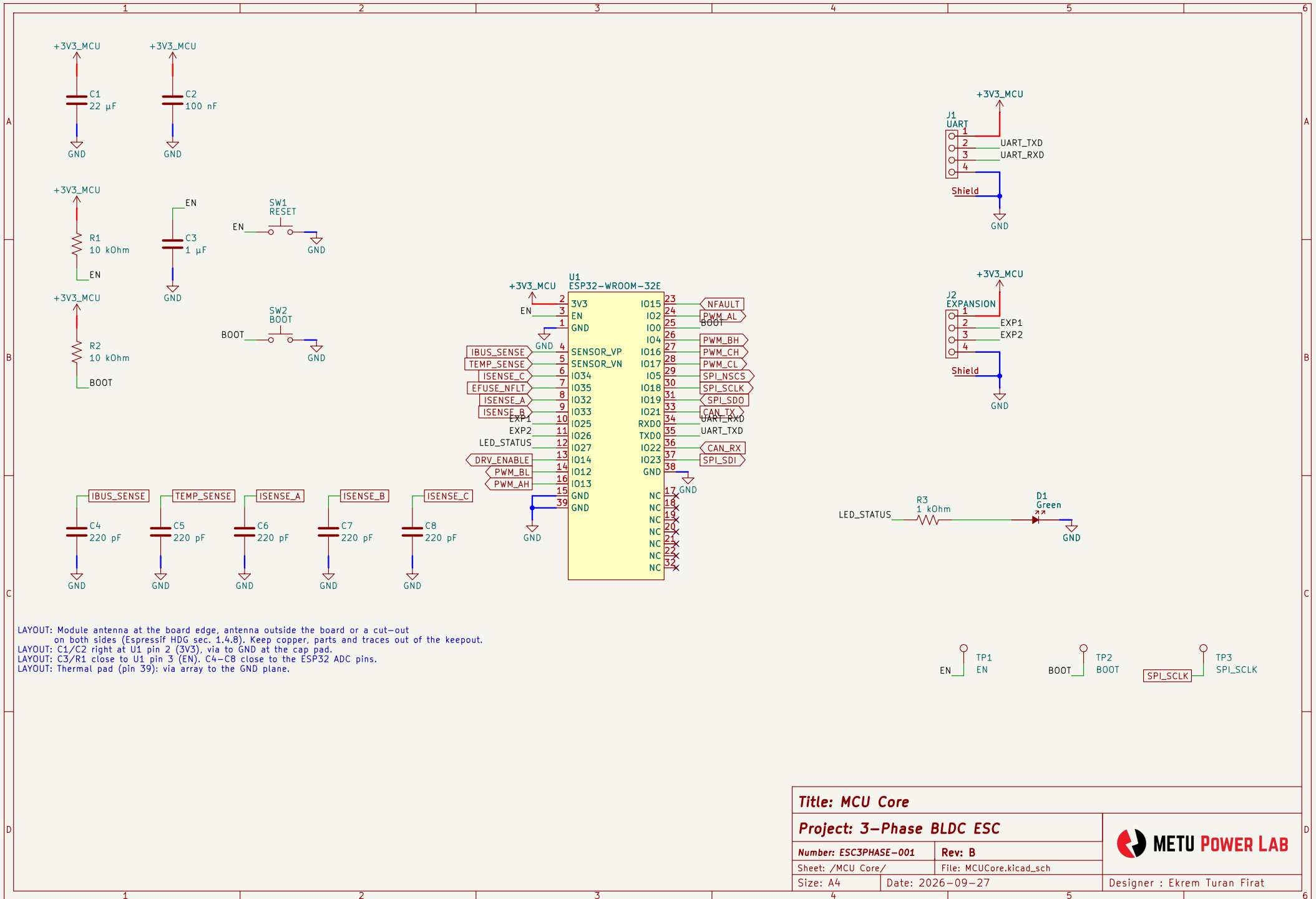


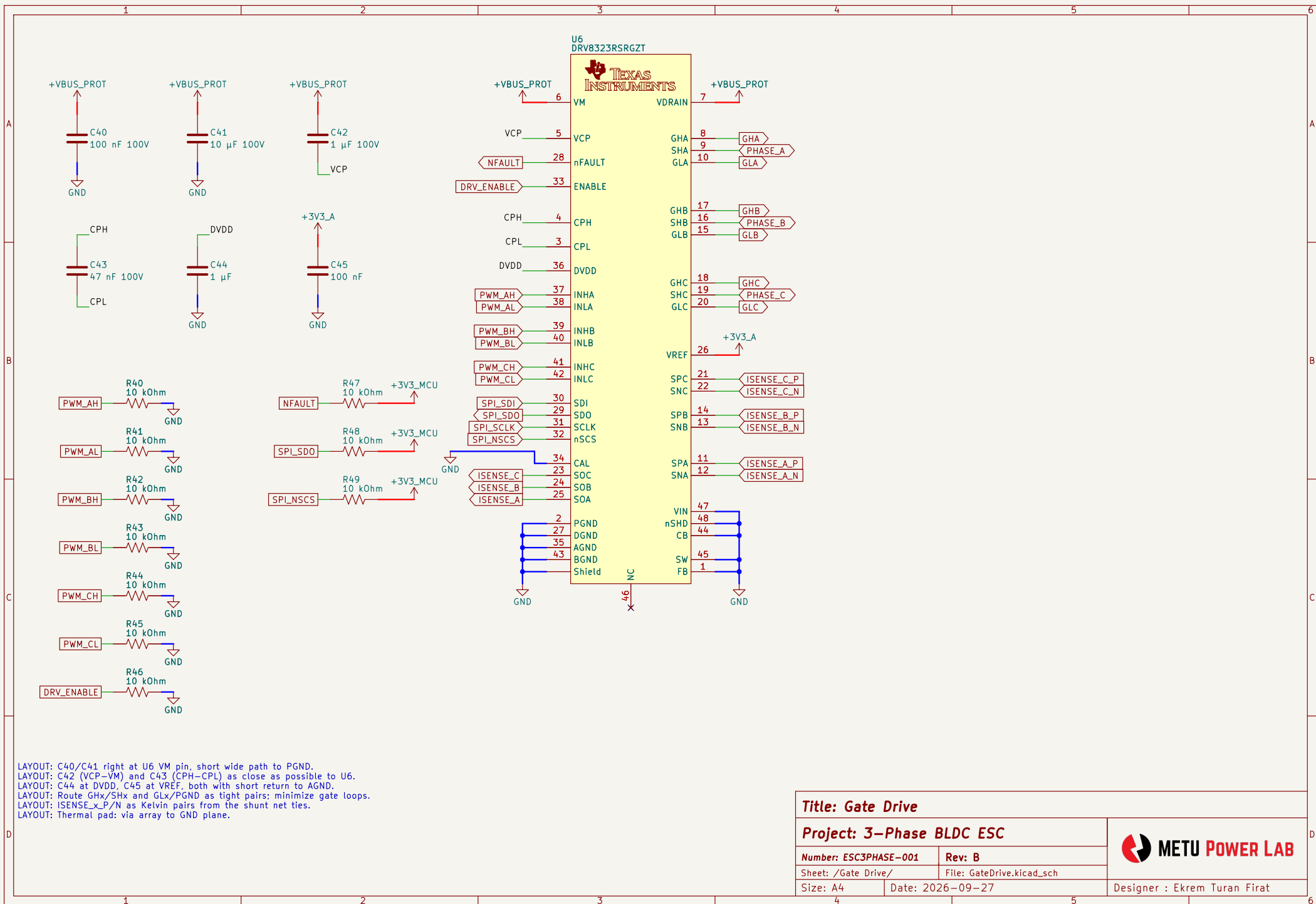
Designer : Ekrem Turan Firat



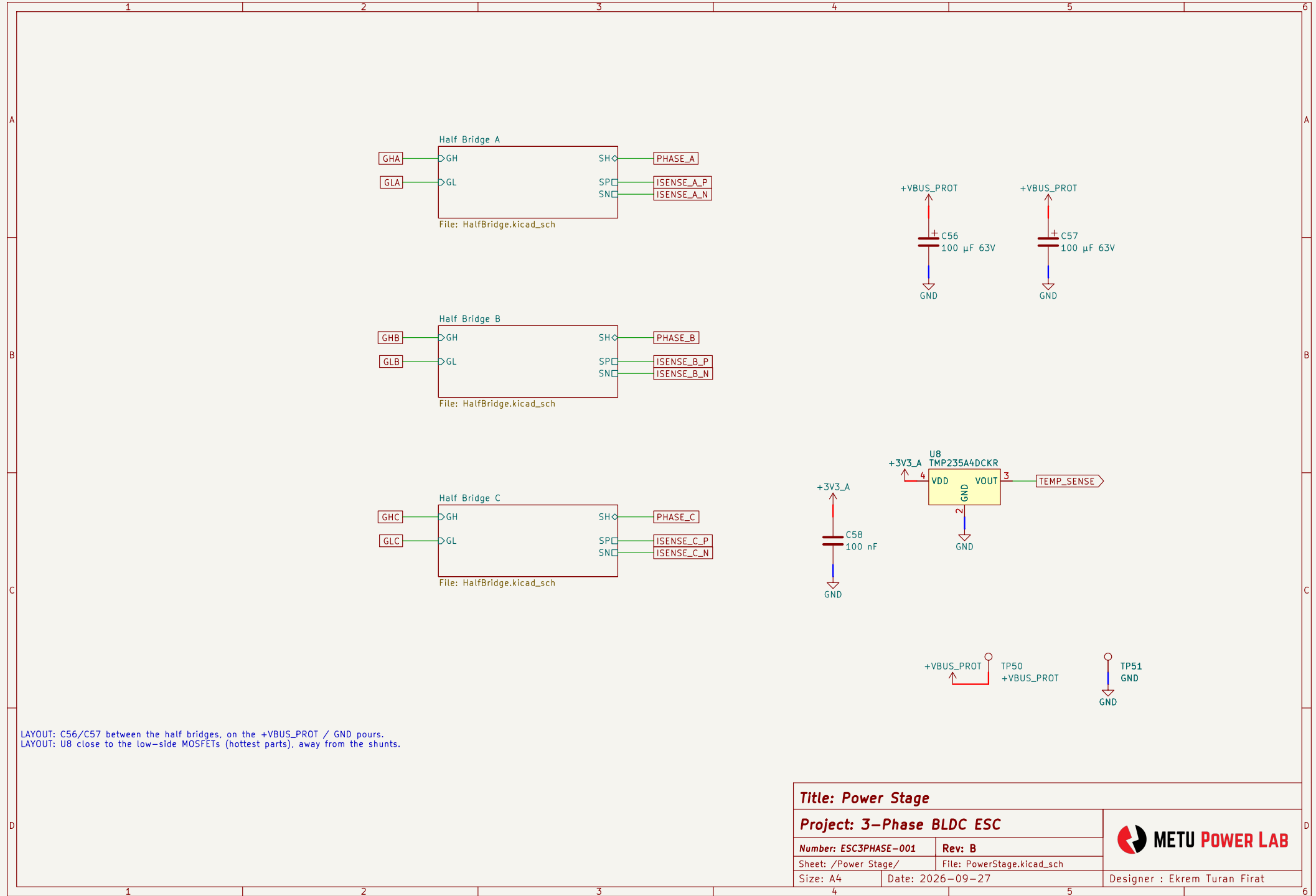
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Project: 3-Phase BLDC ESC	
Number: ESC3PHASE-001	Rev: B
Sheet: /DC-Bus Current Sensing File: CurrentSensing.kicad_sch	
Size: A4	Date: 2026-09-27
Designer : Ekrem Turan Firat	

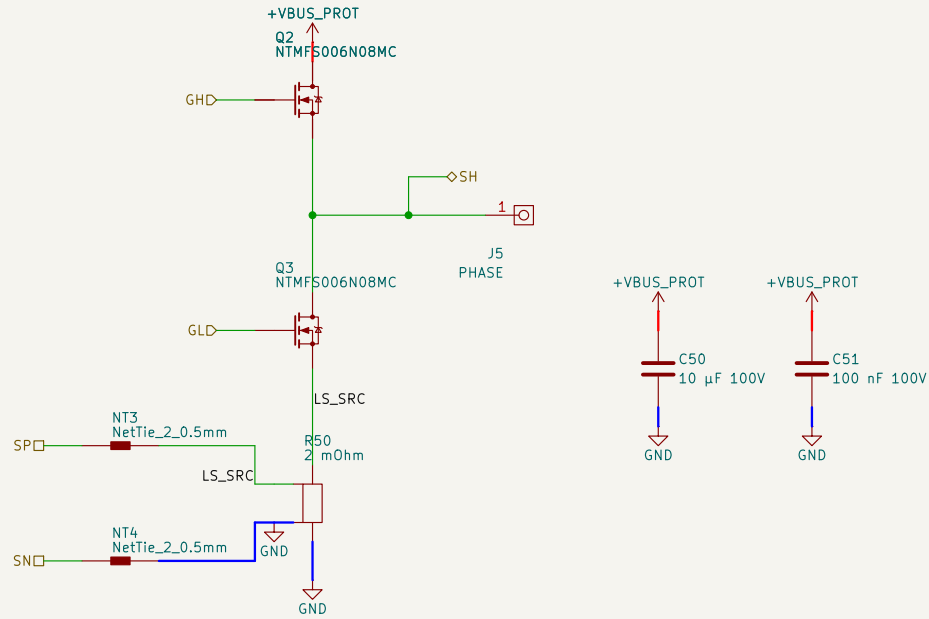






Title: Gate Drive			
Project: 3-Phase BLDC ESC			
Number: ESC3PHASE-001		Rev: B	
Sheet: /Gate Drive/		File: GateDrive.kicad_sch	
Size: A4		Date: 2026-09-27	
		Designer : Ekrem Turan Firat	





LAYOUT: Hot loop: C(10 uF)/C(100 nF) directly across high-side drain and low-side source (shunt GND end); keep this loop as small as possible.
 LAYOUT: PHASE, +VBUS_PROT and GND as pours on both layers with via arrays (5–10 A).
 LAYOUT: Kelvin: net ties at the shunt pads, SP/SN as a tight pair to the DRV8323.
 LAYOUT: Short gate loops: GH with SH, GL with SN/GND return, no layer changes.
 LAYOUT: MOSFET drain pad: thermal vias, copper on both layers.

Title: Half Bridge

Project: 3-Phase BLDC ESC

Number: ESC3PHASE-001

Rev: B

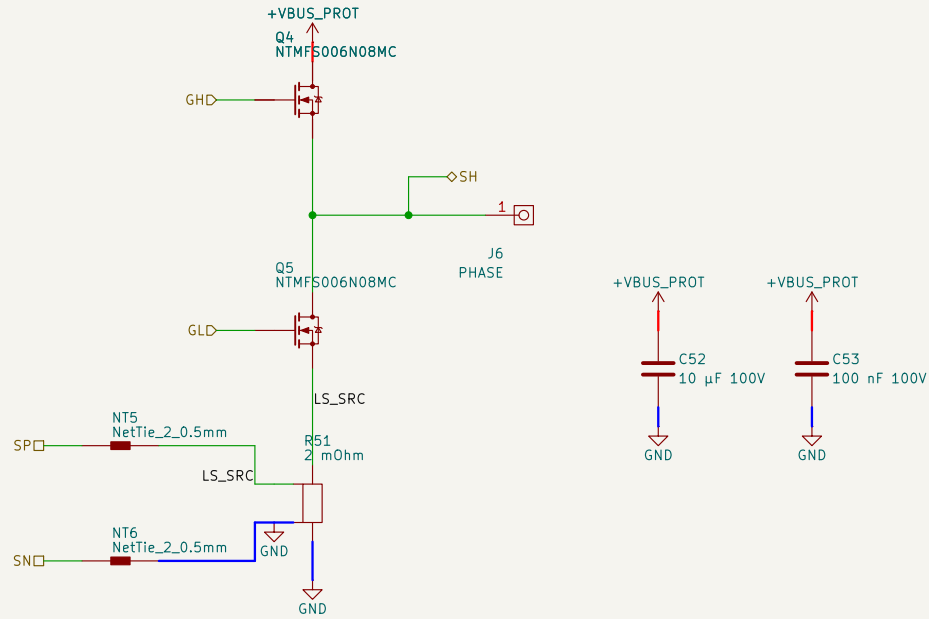
Sheet: /Power Stage/Half Bridge.kicad_sch

Size: A4

Date: 2026-09-27



Designer : Ekrem Turan Firat



LAYOUT: Hot loop: C(10 uF)/C(100 nF) directly across high-side drain and low-side source (shunt GND end); keep this loop as small as possible.
 LAYOUT: PHASE, +VBUS_PROT and GND as pours on both layers with via arrays (5–10 A).
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Title: Half Bridge

Project: 3-Phase BLDC ESC

Number: ESC3PHASE-001

Rev: B

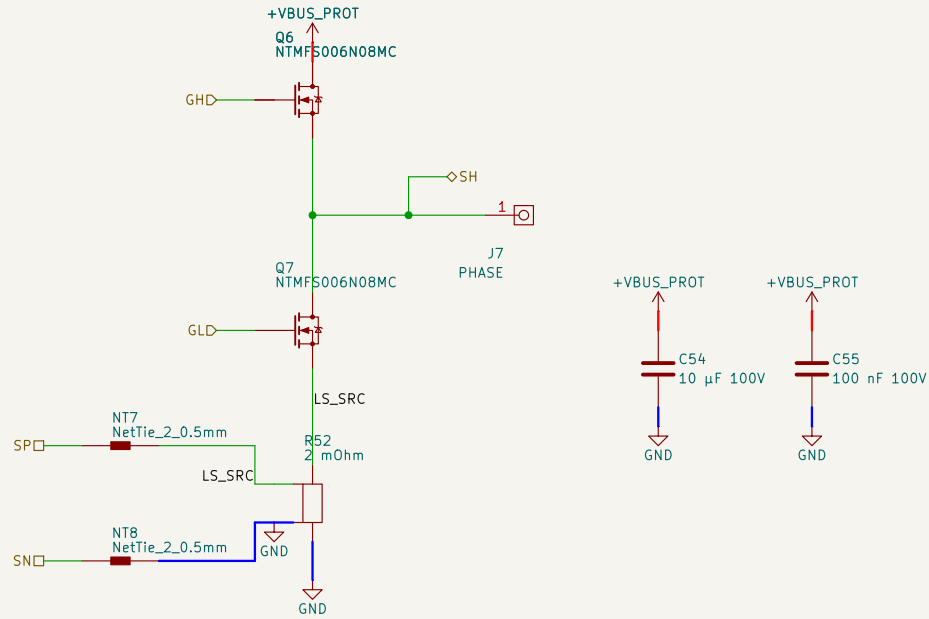
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Size: A4

Date: 2026-09-27



Designer : Ekrem Turan Firat



LAYOUT: Hot loop: C(10 uF)/C(100 nF) directly across high-side drain and low-side source (shunt GND end); keep this loop as small as possible.
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Title: Half Bridge

Project: 3-Phase BLDC ESC

Number: ESC3PHASE-001

Rev: B

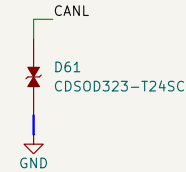
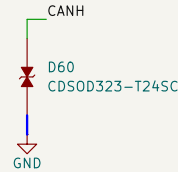
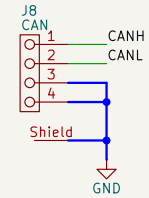
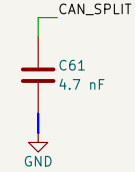
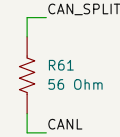
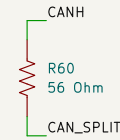
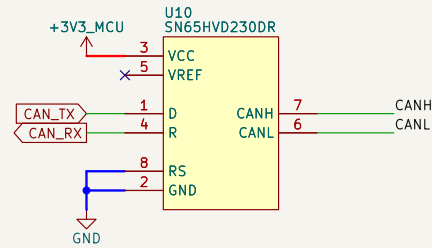
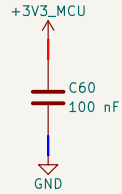
Sheet: /Power Stage/Half Bridge/HalfBridge.kicad_sch

Size: A4

Date: 2026-09-27



Designer : Ekrem Turan Firat



LAYOUT: C60 right at U10 VCC pin.
LAYOUT: CANH/CANL as a tightly coupled pair to J8, same layer, no stubs.
LAYOUT: D60/D61 at the connector, short path to GND. R60/R61/C61 near J8.

Title: Communication

Project: 3-Phase BLDC ESC

Number: ESC3PHASE-001

Rev: B

Sheet: /Communication/

File: Communication.kicad_sch

Size: A4

Date: 2026-09-27



Designer : Ekrem Turan Firat